

AUTONOMOUS ML RESEARCH / TECHJAM 2026

SciOdyssey: Long-Horizon Autonomous ML Research Agent

Autonomous ML discovery through **pure evidence**.

PERSISTENT WORLD

FRESH SCIENTIST

EVIDENCE FIRST

Problem → Motivation → System → Experience → Evaluation → Insights

Can an agent carry a research task to completion?

Turn a task contract into a tested model, a reproducible implementation, and evidence another researcher can inspect.

Explore	Choose hypotheses, write code, and run experiments.	Scientific judgment
Recover	Handle failures and revise unsuccessful ideas.	Long-running work
Deliver	Retain a valid model and the evidence behind it.	Verifiable progress

Our test case: short-video ranking on KuaiRand-Pure, with a fixed evaluator and public-validation boundary.

Why Current Research Agents Fall Short

Three obstacles to sustained autonomous research.

01 CLOSED-WORLD WORKFLOW



Predefined tools and recovery paths cannot handle the unexpected.

Unexpected problems require developer intervention.

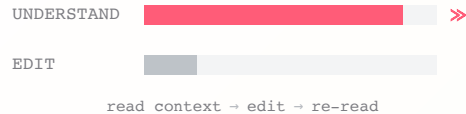
02 SINGLE TRAJECTORY



A long-lived context carries yesterday's momentum into today's search.

More iterations $\neq$ more exploration.

03 TREE SEARCH



Reconstructing file context costs far more tokens than the edit itself.

Context cost forces repeated reading.

Research is an **open-world** task.
We need an **open-world** coding agent.

01 OPEN ACTION SPACE



Shell



Files



Browser



Git

Use what exists

02 SELF-RECOVERY



Read errors



Inspect docs



Inspect
source



Install
packages

Fix what breaks

03 RUNTIME EXTENSION



MCP



Skills



CLI



Packages

Add what's missing

WORKFLOW EXAMPLE · LANGGRAPH

LangGraph makes a known workflow explicit as nodes + edges.

Missing capability? Leave the graph → inspect → extend → retry

Preserve the
research world.

Reset the researcher.

Pass evidence, failures, and code to a fresh Scientist.

Two scientific roles. One safe runtime.

Scientist

Hypothesize · code · test · interpret

One trajectory

META

Audit · preserve · hand off

Across trajectories

Runtime

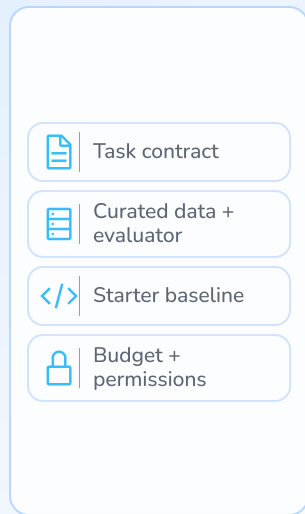
Isolation · cancellation · invariants

System lifetime

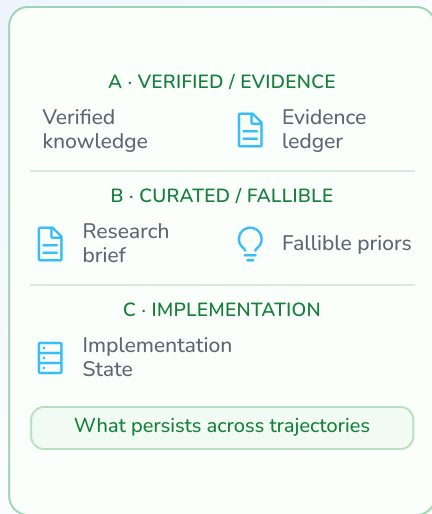
The Scientist chooses the science. META maintains the world. Runtime enforces mechanics, not scientific judgment.

The persistence boundary is explicit.

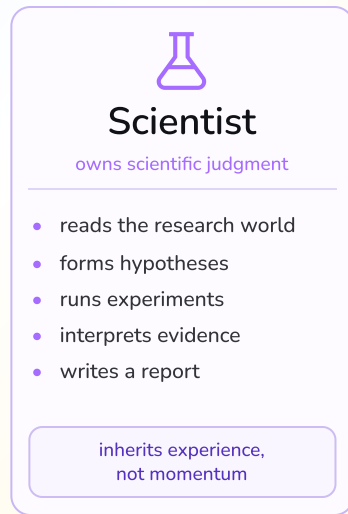
Environment



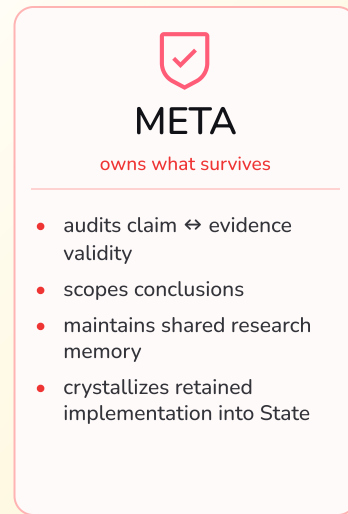
Research World



Fresh Scientist



META Review



Selective persistence / trajectory handoff

audit → scope → preserve

Runtime — deterministic mechanics



Workspace isolation



Runner + evaluator



Logging

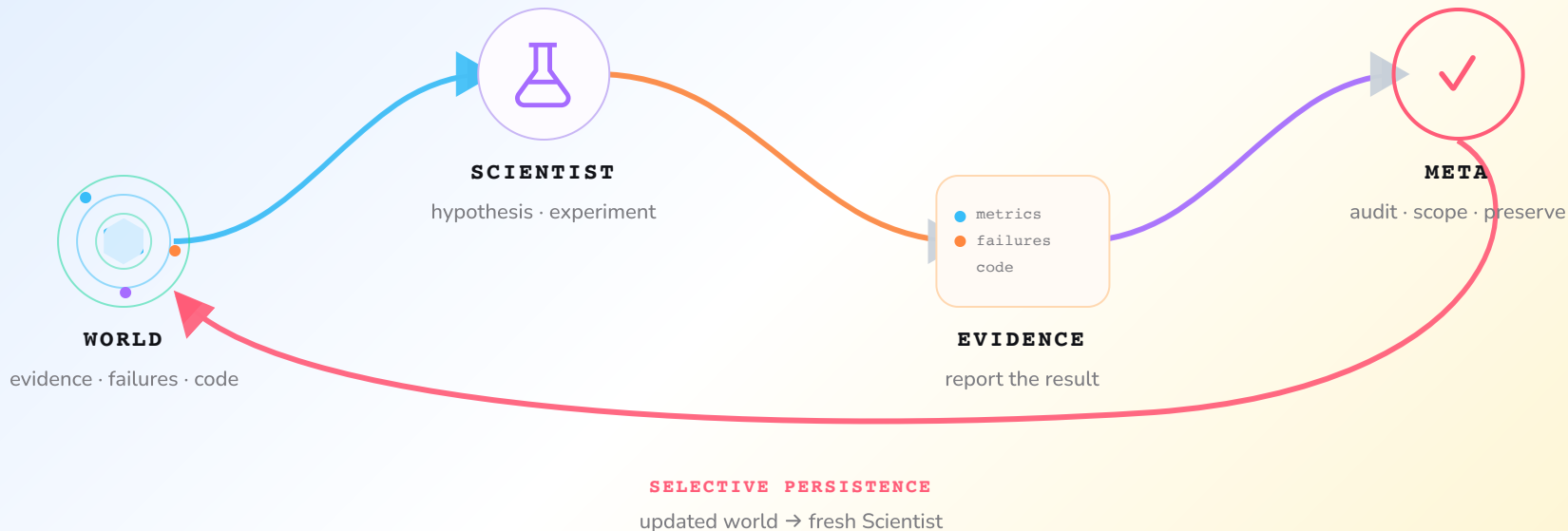


Artifact capture



State operations

Each cycle leaves a world to build on.



PERSISTS metrics · code · failures · original reports

RESTARTS an independent reasoning trajectory

Keep facts distinct from intuition.

research world

- evidence & provenance
- verified knowledge
- current research state
- fallible priors
- State + implementation

Built to be re-examined.

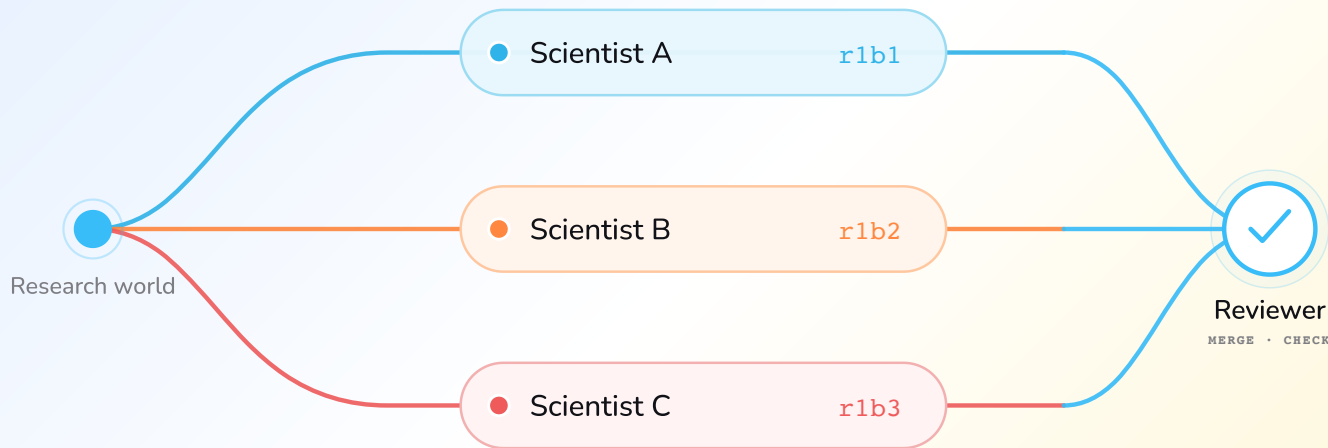
Facts have evidence.

Priors can be overturned.

Code can be picked up again.

Original reports remain available for audit.

One starting point. Independent answers.



Explore in isolated worktrees. Review after completion.

Optional synthesis: one implementation parent + reference evidence from other branches.
Adoption is explicit via parallel-promote. Code and memory are not merged automatically.

Start with the task and working constraints.

The operator defines the objective and constraints; the agent chooses and executes the experiments.

01 task.md

What to solve.

Objectives · constraints · evaluation

EXAMPLE

Rank short videos for each user and interaction context.

KuaiRand-Pure · long_view · GAUC + nDCG@5

02 PERSONAL.md

How you want it solved.

Preferences · research style · priorities

EXAMPLE

Run on macOS / Apple Silicon with uv.

≤ 15 min / experiment · never print credentials

Autonomous by default. Supervisable by design.

Evidence changed the next experiment.

E001–E013: reject weak ideas, improve representations, then test ensembles.

E001	Screen historical target encodings; reject weak variants.	Medium screening
E003	Establish a valid rich-FM checkpoint.	Full · 0.6016310
E008	Retain a five-seed, 38-field FM ensemble.	Full · 0.6040901
E013	Select the eight-seed, 46-field FM ensemble.	Full · 0.6059363

These are milestones within four autonomous cycles. Medium screens ideas; Full evaluations support retained score claims.

An evaluation failure became a recoverable step.

Cycle 1 completed training and evaluation, then failed while writing the evidence file.

Failure

NumPy float32 values broke JSON serialization.

Evidence writer

Repair

The Scientist converted scalars and reran.

Agent-authored fix

Retain

Printed measurements and saved evidence survived.

Auditable recovery

No human selected this repair or edited the code. This is one observed recovery case, not a guarantee for every failure.

A retained state keeps the evidence inspectable.

The screenshot displays a 'Research Dashboard' interface. At the top right, it says 'RECORDED VALIDATION'. The main content area features a 'Retained Validation Result' section for 'S004 · RETAINED STATE'. This section is highlighted with a green border. Inside, it lists 'SCORE-BEARING CHECKPOINT S004' and describes an '8-seed ensemble of 46-field Factorization Machine rankers'. Below this, a table of metrics is shown: GAUC (0.672842), nDCG@5 (0.539030), Primary (0.605936), and Delta vs baseline (+0.004336). The evidence path is listed as 'system/evidence/cycle4-fm-rich46-ensemble8-full.json'. At the bottom, there are three buttons: 'Research Intuition', 'Recent Activity', and 'Run Summary'.

GAUC	nDCG@5	Primary	Delta vs baseline
0.672842	0.539030	0.605936	+0.004336

Evidence · system/evidence/cycle4-fm-rich46-ensemble8-full.json

Inspect the retained state and the evidence behind it.

Retained validation checkpoint · evidence path · reproducible implementation.

Autonomous research. Measured gains.

PRIMARY / RESEARCH AGENT

0.6059363

+0.0043363 absolute improvement

METRIC	Official FM	Ours
GAUC	0.6674000	0.6728421
nDCG@5	0.5357000	0.5390304
Primary	0.6016000	0.6059363

Primary = mean(GAUC, nDCG@5). Public validation only; hidden-test scoring is organizer-controlled.

Final recipe: 46 categorical features × 8-seed FM ensemble · NumPy / CPU

An audit trail from experiment to output.

4

Autonomous cycles

Within a 50-iteration cap

13

Named experiments

E001–E013 · 7 Full evaluations

0

Manual scientific interventions

After launch · 0 GPU-hours

170,588 prediction rows passed the unchanged Starter Kit alignment checker.

LLM tokens: 48.24M including cache reads; 4.02M excluding cache reads.

Resource use is part of the evidence.

Project-level subscription estimate, with run-level telemetry reported separately.

~\$10

from first line of code to final result

coding · debugging · agent runs · all experiments

MODEL

One GPT Plus + One Gemini Pro for 7 days

Two consumer subscriptions covered the entire research loop.

COMPUTE

1 × MacBook M2

All development and experiments ran on a single consumer laptop.

The recorded runs used CPU training; LLM usage remains a separate cost.

Compare scores alongside their evidence.



Runs differ in model, budget, and evidence quality; this is not a controlled causal comparison.

The run taught us more than the final score.

What survived the experiments — and what remains unsettled.

01 Diversity is a prior.

Keep `gemini-3.7-flash` as META while Scientist rotates across `gpt-5.6-sol`, `gemini-3.7-flash`, and `gpt-5.6-luna`. This widens priors; it does not prove the gain.

03 Failure can be evidence.

After evaluation, NumPy `float32` values broke serialization. The Scientist fixed the writer, reran, and kept the measurements.

02 A score is not a claim.

Audits found leakage and sampling mismatches in competing experiments. META can scope evidence; it is not a formal verifier.

04 Reset $\neq$ no anchor.

Fresh cognition reopens the search, but shared summaries can still omit context or anchor future work. Share evidence later than cognition.

OBSERVED, NOT PROVEN

Parallel/Synthesis reached public-validation Primary 0.6055536; the canonical E001–E013 frontier remains 0.6059363.

4 cycles

0 post-launch interventions

0 GPU-hours

Reproducible evidence. Open questions.

01	One task does not establish generality.	External validity
02	META, State, and reset are not isolated.	Causal attribution
03	Summaries can lose context and anchor.	Memory quality

Next questions: broader tasks, targeted ablations, and less repeated execution.

A clean repo. A continuous agent.

FIELD NOTE / 01

Engineering practices to apply next: isolate parallel changes and make unattended work reviewable.

01 MANY AGENTS



Give every agent a branch.

main

clean trunk

● agent A / branch

● agent B / branch

● agent C / branch

review

→ merge

Let agents work independently. Conflicts stay local, and the main branch stays clean.

02 NO QUOTA / NO LAPTOP



Leave the next move in GitHub.

01

Issue /
prompt

inject the task



02

GitHub
Action

run the agent



03

Commit

review the change

Connect the agent to GitHub Actions. It can code and commit while you are offline; review before merging.

Keep the experience. Reopen the search.

```
# From the configured repo, run one autonomous cycle
$ ./scripts/research-agent step --cli codex \
    --target /absolute/path/to/project --allow-edits
```

github.com/zc6600/research-agent-kuairand ↗

Code · Technical report · Experiment ledger · Checked output

Five people. One autonomous research loop.

OVERLAPPING RESPONSIBILITIES / SHARED WORLD



Chen Zhu

TEAM LEAD / SYSTEM

Direction · architecture ·
integration



Zhou Ziyu

MODELING / EXPERIMENTS

Features · ensembles ·
evidence review



Shilin Xu

DATA / METRICS

Data workflow · contracts
· alignment



GE GAO

RUNTIME / RELIABILITY

Execution · testing ·
integration support



Jiran Li

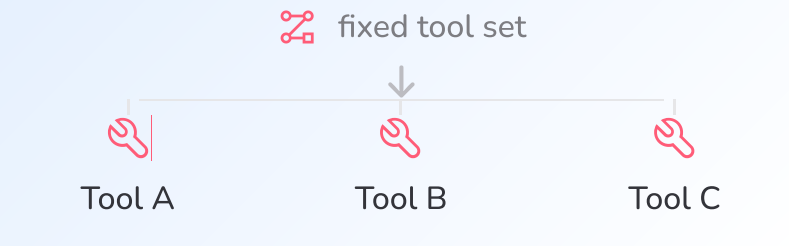
EVIDENCE /
REPRODUCIBILITY

Telemetry · documentation
· packaging

The difference is not orchestration. It is recoverability.

LANGGRAPH-STYLE ORCHESTRATION

predefined graph

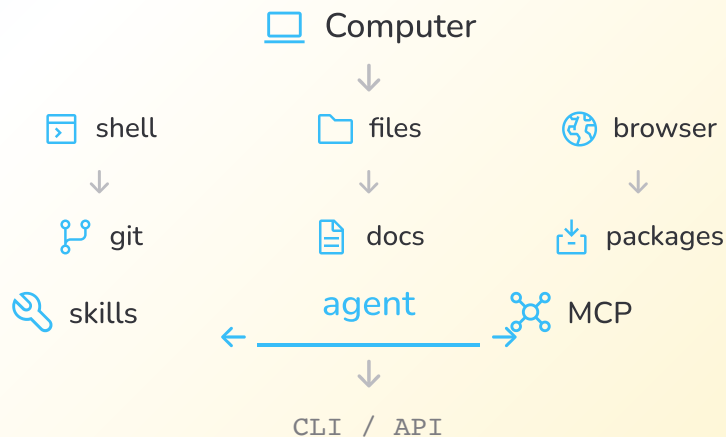


- ⊘ API fails
- ↓
- ⊘ no recovery tool
- ↓
- ⊘ **DEAD END**

Capabilities are bounded by what developers anticipated.

CODING-AGENT HARNESS

open action space



fail → inspect → learn → modify environment → retry

The agent can acquire the missing capability at runtime.